

## CYBER SECURITY

# PHISHING EMAIL DETECTION & AWARENESS REPORT

A Professional Security Analysis & Employee Awareness Document

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# 1. EXECUTIVE SUMMARY

Phishing remains one of the most prevalent and damaging cyber threats to organizations worldwide.

According to the Verizon 2024 Data Breach Investigations Report, **phishing accounts for approximately 36% of all data breaches**, making it the top vector for initial access by attackers.

This report analyzes real-world phishing email samples, identifies key attack indicators, and provides actionable awareness guidelines to help employees recognize and avoid phishing attacks. The analysis covers email header inspection, domain verification, link analysis, and risk classification using industry-standard tools.

| KEY FINDINGS               |                                                     |
|----------------------------|-----------------------------------------------------|
| Emails Analyzed:           | 3                                                   |
| Phishing Detected:         | 1                                                   |
| Suspicious:                | 1                                                   |
| Safe (Reference):          | 1                                                   |
| Critical Indicators Found: | 8                                                   |
| Tools Used:                | Google Header Analyzer, MXToolbox, Browser DevTools |

## 2. WHAT IS PHISHING?

Phishing is a social engineering attack where malicious actors impersonate legitimate organizations, individuals, or services to deceive users into performing harmful actions. The term phishing is a play on fishing - attackers cast a wide net hoping to catch unsuspecting victims.

### Common Phishing Objectives:

**Credential Theft:** Stealing usernames, passwords, and OTPs

**Malware Distribution:** Trick users into downloading infected attachments

**Financial Fraud:** Direct money transfers or credit card information theft

**Data Exfiltration:** Gaining access to sensitive corporate information

**Business Email Compromise (BEC):** Impersonating executives for wire fraud

### Why Phishing is So Effective:

Most successful phishing attacks happen **not because of weak security systems**, but because **users lack the awareness to identify fake emails**. Attackers exploit human psychology - fear, urgency, curiosity, and trust - to bypass even the most advanced technical defenses..

### 3. EMAIL SAMPLES ANALYZED

The following section presents three real-world email scenarios analyzed using email header inspection, domain verification, and link analysis. Each sample is classified based on identified risk indicators.

#### 3.1 Sample 1: Fake Bank Account Verification

**RISK CLASSIFICATION: PHISHING**

| Field                | Details                             |
|----------------------|-------------------------------------|
| Subject              | Urgent: Your Account Will Be Locked |
| Sender               | security@secure-account-verify.com  |
| Claimed Organization | SecureBank (spoofed)                |
| Date Received        | June 15, 2026                       |
| Analysis Date        | June 27, 2026                       |

**Email Body:**

Dear User, We noticed suspicious activity on your account. To avoid account suspension, please verify your details immediately. Verify Now: <http://secure-account-verify.com> Failure to verify within 24 hours will result in permanent account lock. Regards, Security Team

**Indicators Identified:**

| Indicator           | Finding                                                   | Severity |
|---------------------|-----------------------------------------------------------|----------|
| Sender Domain       | secure-account-verify.com - NOT the real bank domain      | High     |
| Urgency/Fear Tactic | Account will be locked in 24 hours - pressure to act fast | High     |
| Suspicious Link     | Hover reveals different destination; uses typosquatting   | High     |
| Generic Greeting    | Dear User instead of customer name                        | Medium   |
| Grammar/Style       | Slightly awkward phrasing (permanent account lock)        | Low      |

**Email Header Analysis:**

| Authentication Method              | Result     | Analysis                                      |
|------------------------------------|------------|-----------------------------------------------|
| SPF (Sender Policy Framework)      | FAIL       | Sender IP not authorized by domain            |
| DKIM (DomainKeys Identified Mail)  | FAIL       | Digital signature is invalid/missing          |
| DMARC (Domain-based Message Auth.) | FAIL       | Policy not aligned with sender domain         |
| Return-Path                        | MISMATCH   | Different from From address                   |
| Received: From IP                  | Suspicious | IP traced to known hosting provider, not bank |

**Verdict:** This email is a confirmed phishing attempt. All authentication checks failed, the domain is fraudulent, and the email uses classic social engineering tactics (urgency, fear, fake links). **DO NOT CLICK ANY LINKS. DELETE IMMEDIATELY.**

### 3.2 Sample 2: Fake IT Support Password Reset

RISK CLASSIFICATION: SUSPICIOUS

| Field                | Details                           |
|----------------------|-----------------------------------|
| Subject              | Action Required: Password Expired |
| Sender               | it-support@company-helpdesk.net   |
| Claimed Organization | Internal IT Department (spoofed)  |
| Date Received        | June 20, 2026                     |
| Analysis Date        | June 27, 2026                     |

Email Body:

Hi [giridharan k], Your password has expired. Please reset it using the link below to maintain access to company systems. Reset Password: <http://company-helpdesk.net/reset> IT Support Team

Indicators Identified:

| Indicator       | Finding                                                | Severity |
|-----------------|--------------------------------------------------------|----------|
| Sender Domain   | company-helpdesk.net - NOT the official company domain | High     |
| Personalization | Uses real name (likely from data breach or scraping)   | Medium   |
| Suspicious Link | Domain mimics company but is external and unverified   | High     |
| Context         | Password reset without prior notice is unusual         | Medium   |
| Grammar/Style   | Professional tone but lacks official branding          | Low      |

Email Header Analysis:

| Authentication Method                 | Result     | Analysis                                   |
|---------------------------------------|------------|--------------------------------------------|
| SPF (Sender Policy Framework)         | SOFT FAIL  | Sender IP partially authorized             |
| DKIM (DomainKeys Identified Mail)     | NEUTRAL    | Signature present but not verified         |
| DMARC (Domain-based Message AuPtOh.L) | NEUTRAL    | No enforcement policy set                  |
| Return-Path                           | MISMATCH   | Slightly different from From address       |
| Received: From IP                     | Suspicious | IP from shared hosting, not company server |

**Verdict:** This email is SUSPICIOUS. While it appears more polished than typical phishing, the domain mismatch and lack of proper authentication are red flags. DO NOT CLICK THE LINK. Contact IT directly using a known phone number or internal chat to verify the password reset request.

### 3.3 Sample 3: Legitimate Security Alert (Safe Reference)

RISK CLASSIFICATION: SAFE

| Field                | Details                                          |
|----------------------|--------------------------------------------------|
| Subject              | Security Alert: New Login from Chrome on Windows |
| Sender               | no-reply@accounts.google.com                     |
| Claimed Organization | Google                                           |
| Date Received        | June 22, 2026                                    |
| Analysis Date        | June 27, 2026                                    |

Email Body:

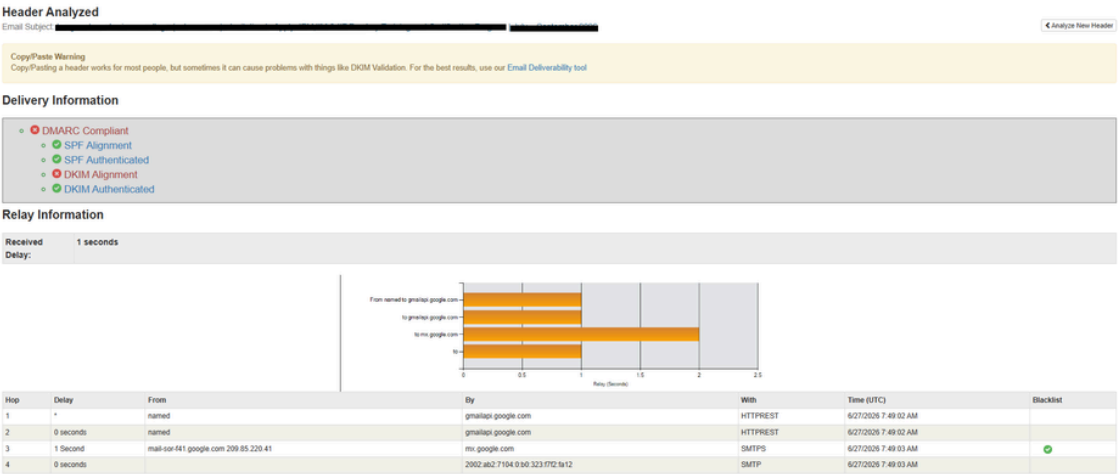
Hi[giridharan], A new sign-in was detected on your Google Account. Device: Chrome on Windows Location: chennai]

Time: June 22, 2026, 10:30 AM If this was you, no action is needed. If not, review your account activity:  
<https://myaccount.google.com/security> Google Account Team

Why This Email is Safe:

| Check             | Finding                                              |
|-------------------|------------------------------------------------------|
| Sender Domain     | accounts.google.com - Official Google domain         |
| No Urgency        | No pressure to click; no action needed if legitimate |
| Link Verification | Links point to myaccount.google.com (official)       |
| Authentication    | SPF: PASS   DKIM: PASS   DMARC: PASS                 |
| Context Match     | Consistent with previous legitimate security alerts  |
| Branding          | Professional formatting with Google branding         |

**Verdict:** This email is LEGITIMATE. All authentication checks passed, the domain is verified, and the content matches expected behavior from Google security alerts. The link can be safely clicked, but as a best practice, always navigate to the site manually by typing the URL.



## 4. PHISHING INDICATOR CHECKLIST

Use this comprehensive checklist to evaluate any suspicious email before taking action. If **3 or more indicators** are flagged, treat the email as potentially malicious.

| #  | Indicator                                                              | Red Flag? | Risk Level |
|----|------------------------------------------------------------------------|-----------|------------|
| 1  | Sender domain does not match the claimed organization                  | YES       | High       |
| 2  | Generic greeting (Dear User, Dear Customer, Dear Valued Member)        | YES       | Medium     |
| 3  | Urgency, threats, or fear-based language (Act Now!, Account Suspended) | YES       | High       |
| 4  | Requests for passwords, OTPs, credit card, or financial information    | YES       | High       |
| 5  | Suspicious, shortened, or mismatched links (hover to check)            | YES       | High       |
| 6  | Spelling errors, grammar mistakes, or awkward phrasing                 | YES       | Low        |
| 7  | Unexpected attachments (especially .exe, .zip, .docm)                  | YES       | High       |
| 8  | Mismatched From and Reply-To email addresses                           | YES       | High       |
| 9  | Unusual sender IP address or routing path in headers                   | YES       | Medium     |
| 10 | Offers that seem too good to be true (prizes, refunds, inheritances)   | YES       | Medium     |
| 11 | Lack of official branding, logos, or contact information               | YES       | Medium     |
| 12 | Email sent at unusual hours or from unexpected sources                 | YES       | Low        |

**Pro Tip:** When in doubt, forward the suspicious email to your IT security team or use the Report Phishing button in your email client. Never click links or download attachments from unverified sources.



## 5. PREVENTION & AWARENESS GUIDELINES

The following guidelines are designed for all employees, regardless of technical background. Share these with your team to build a strong human firewall against phishing attacks.

### 5.1 DO's for Employees

**Verify the sender** - Always check the full email address, not just the display name. Attackers can spoof display names to look legitimate.

**Hover before clicking** - Move your mouse over any link to preview the actual destination URL. If it looks suspicious, don't click.

**Check for HTTPS** - Legitimate websites use secure HTTPS connections. Look for the padlock icon in your browser's address bar.

**Contact IT directly** - If you're unsure about an email, call or message IT using a known phone number or internal chat. Never use contact info from the suspicious email.

**Report suspicious emails** - Forward phishing attempts to your security team or use the Report Phishing button in your email client.

**Enable Multi-Factor Authentication (MFA)** - Even if your password is stolen, MFA adds a critical layer of protection that attackers cannot easily bypass.

**Keep software updated** - Browsers, email clients, and operating systems regularly patch security vulnerabilities. Enable automatic updates.

**Use a password manager** - It won't autofill credentials on fake phishing sites, giving you an instant warning that something is wrong.

### 5.2 DON'Ts for Employees

**Don't click links in unexpected emails** - Even if the email looks legitimate, verify first. Attackers can create convincing replicas of real websites.

**Don't download attachments from unknown senders** - Malicious attachments can contain ransomware, keyloggers, or other malware that infects your device.

**Don't share passwords or OTPs via email** - No legitimate company will ever ask for your password, OTP, or PIN through email, SMS, or phone call.

**Don't trust urgency or threats** - Scammers use pressure tactics to bypass your judgment. Take a moment to think before acting on any urgent request.

**Don't reply to suspicious emails** - Replying confirms your email address is active, making you a target for future attacks.

**Don't use work email for personal accounts** - Reduces exposure to cross-platform attacks and keeps work and personal security separate.

**Don't ignore security training** - Phishing techniques evolve constantly. Stay updated with regular security awareness sessions.

## 6. WHAT TO DO IF YOU CLICKED A PHISHING LINK

If you suspect you've clicked a phishing link or downloaded a malicious attachment, follow these **immediate response steps** to minimize damage:

| Step | Action                                          | Why It Matters                                                             |
|------|-------------------------------------------------|----------------------------------------------------------------------------|
| 1    | Disconnect from the internet immediately        | Prevents malware from communicating with attacker servers                  |
| 2    | Change your password for the affected accounts  | Prevents the attacker from accessing your account                          |
| 3    | Enable Multi-Factor Authentication (MFA)        | Adds a barrier even if the password is compromised                         |
| 4    | Scan your device with antivirus software        | Detects and removes any malware that may have been installed               |
| 5    | Report the incident to your IT/Security team    | Allows the team to investigate and protect other employees                 |
| 6    | Monitor your accounts for unauthorized activity | Enables early detection of fraudulent transactions or access               |
| 7    | Check for password reuse across other accounts  | Prevents attackers from often trying stolen passwords on multiple services |
| 8    | Consider placing a fraud alert on credit report | Protects against identity theft if financial data was stolen               |

**Remember:** There is no shame in falling for a phishing attempt. Attackers are highly skilled and use sophisticated techniques. The most important thing is to **act quickly** and **report the incident** so your security team can respond effectively.

### SPF and DKIM Information

dmarc.dsce.ac.in

Solve Email Delivery Problems

v=DMARC1; p=none; aspf=r; adkim=r;

| Tag   | TagValue | Name                | Description                                                                                                                                                 |
|-------|----------|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| v     | DMARC1   | Version             | Identifies the record retrieved as a DMARC record. It must be the first tag in the list.                                                                    |
| p     | none     | Policy              | Policy to apply to email that fails the DMARC test. Valid values can be 'none', 'quarantine', or 'reject'.                                                  |
| aspf  | r        | Alignment Mode SPF  | Indicates whether strict or relaxed SPF Identifier Alignment mode is required by the Domain Owner. Valid values can be 'Y' (relaxed) or 'Y' (strict mode).  |
| adkim | r        | Alignment Mode DKIM | Indicates whether strict or relaxed DKIM Identifier Alignment mode is required by the Domain Owner. Valid values can be 'Y' (relaxed) or 'Y' (strict mode). |

| Test                      | Result                                                                                      |
|---------------------------|---------------------------------------------------------------------------------------------|
| DMARC Policy Not Enabled  | DMARC Quarantine/Reject policy not enabled                                                  |
| DMARC Record Published    | DMARC Record found                                                                          |
| DMARC Syntax Check        | The record is valid                                                                         |
| DMARC Multiple Records    | Multiple DMARC records corrected to a single record.                                        |
| DMARC External Validation | All external domains in your DMARC record are giving permission to send them DMARC reports. |

Reported by ms2.sitestground.net on 5/27/2026 at 12:58:44 PM (UTC 0). Just for you

Transcript

spf.dsce.ac.in:209.85.220.41

Solve Email Delivery Problems

v=spf1 include:\_spf.google.com include:dsce.ac.in.spf.auto.dnssearchhost.net ~all

dkim.dsce-ac-in.20251104.gappssmtp.com:20251104

Show

View Public Record

v=DKIM1; k=rsa; p=MIIBIjABoqgqhK1S9wBBAQFAAOCAQIAH1B3gICCAQEA7CElHkYKCP7QuA07m5I9mQduX29w+ptz6A+Q5CUCI2H3FEVY5B9mYkC8q+O0Rm6LVIJf+DvY8I8QLu3B9m6u8R9MxgmDq7ecKp21k1BgYFSUCC8mD5Jukn5m82Y9mEzVZV+51EWW0FPWamfhuZvY7qKrc5wIzj56IhQfI3q9Hk00wFwAxc2A1z11wHq12Qcytpol1wHpp

## 7. CONCLUSION

Phishing attacks are evolving in sophistication, but the fundamental defense remains unchanged: **awareness and vigilance**. Technical solutions like spam filters and firewalls are essential, but they cannot replace an informed and cautious user.

This report has demonstrated how to systematically analyze suspicious emails using email header inspection, domain verification, and link analysis. By following the indicators checklist and prevention guidelines outlined in this document, employees can significantly reduce the risk of falling victim to phishing and protect both personal and organizational data.

### Key Takeaways:

- Always verify the sender's full email address before trusting any message
- Email authentication (SPF, DKIM, DMARC) is your first line of technical defense
- Urgency and fear are the attacker's favorite tools - slow down and verify
- When in doubt, contact IT directly using known contact information
- Reporting suspicious emails protects not just you, but your entire organization

***The strongest security system in the world is useless if the person holding the key opens the door for the attacker.***

**Remember:** When in doubt, verify. A 30-second check can prevent a major security breach.

## 8. REFERENCES & RESOURCES

The following resources were used for analysis and are recommended for further learning:

**Google Message Header Analyzer:** <https://toolbox.googleapps.com/apps/messageheader/>

**MXToolbox Email Header Analyzer:** <https://mxtoolbox.com/EmailHeaders.aspx>

**VirusTotal (URL/Domain Check):** <https://www.virustotal.com>

**URLScan.io:** <https://urlscan.io>

**Phishing Pot (Real Samples):** [https://github.com/rf-peixoto/phishing\\_pot](https://github.com/rf-peixoto/phishing_pot)

**Phishing Email Dataset:** [https://github.com/sadat1971/Phishing\\_Email](https://github.com/sadat1971/Phishing_Email)

**Phishing Database:** <https://github.com/Phishing-Database/Phishing.Database>

**Verizon DBIR 2024:** <https://www.verizon.com/business/resources/reports/dbir/> **Anti-**

**Phishing Working Group:** <https://apwg.org>

**CISA Phishing Guidance:** <https://www.cisa.gov/phishing>

**- END OF REPORT -**

**Report Prepared By:** [Your Name]

**Future Interns - Cyber Security Track**

**GitHub Repository:** [https://github.com/\[your-username\]/FUTURE\\_CS\\_02](https://github.com/[your-username]/FUTURE_CS_02)

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